Source Separation 2

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Speech And Audio Recognition
(오디오 음성인식)

https://mairlab-km.github.io/

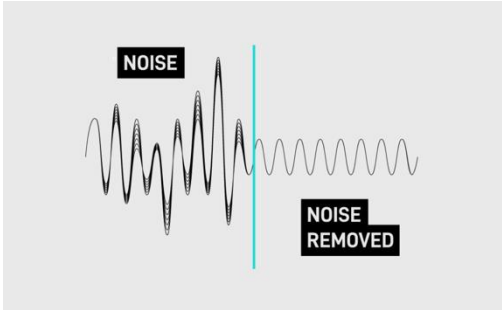
This lecture material refers to
https://github.com/yandexdataschool/speech_course?tab=readme-ov-file and
https://github.com/markovka17/dla

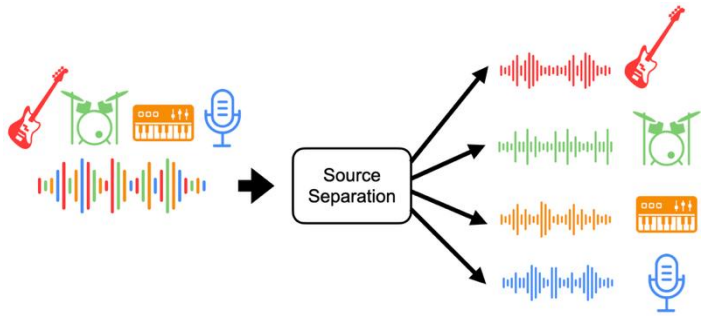
What is Source Separation?

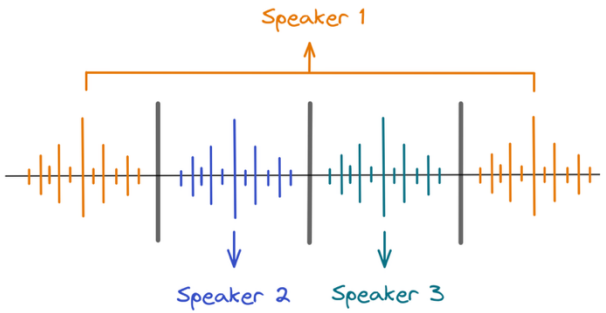
- Source Separation literally means separate any source of particular interest…

Separation?

- Separate signal (music, speech, …) from noise → **Denoising**
- Separate audio mix into separate tracks → **Source Separation**
- Guess when specific sound sources are active → **Diarization**





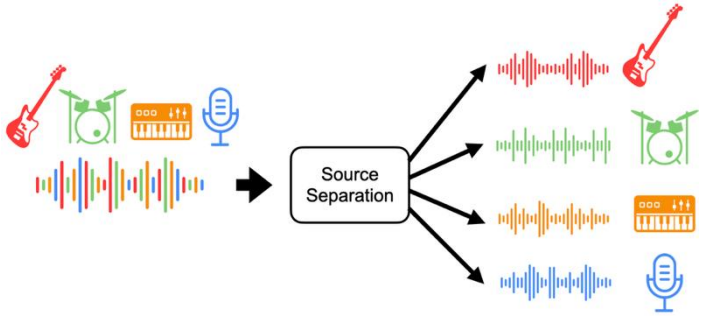


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Blind (any k)

Guided (specifically bass, vocal, guitar)

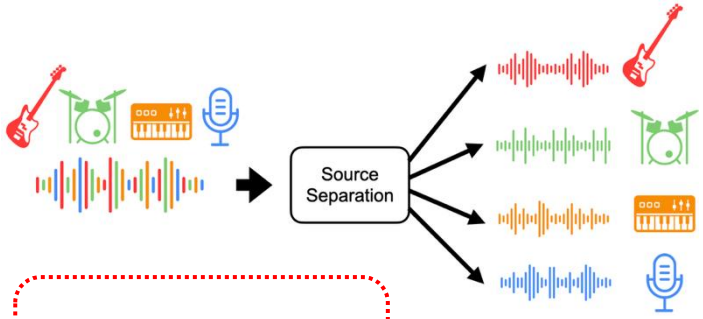
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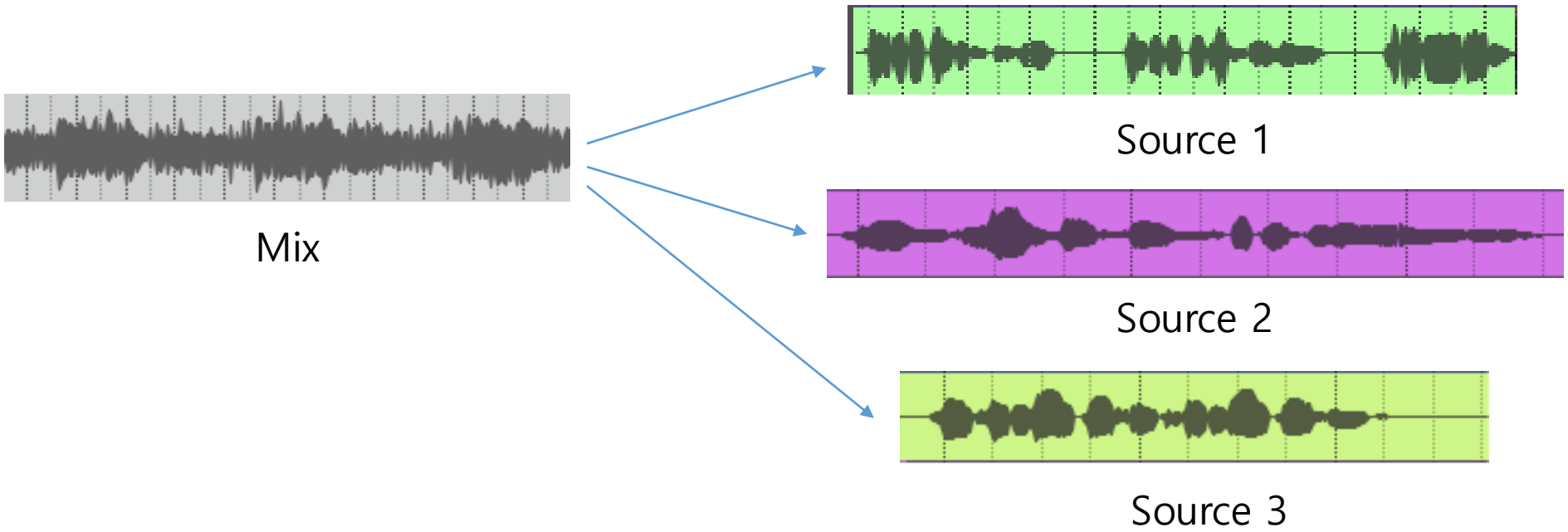
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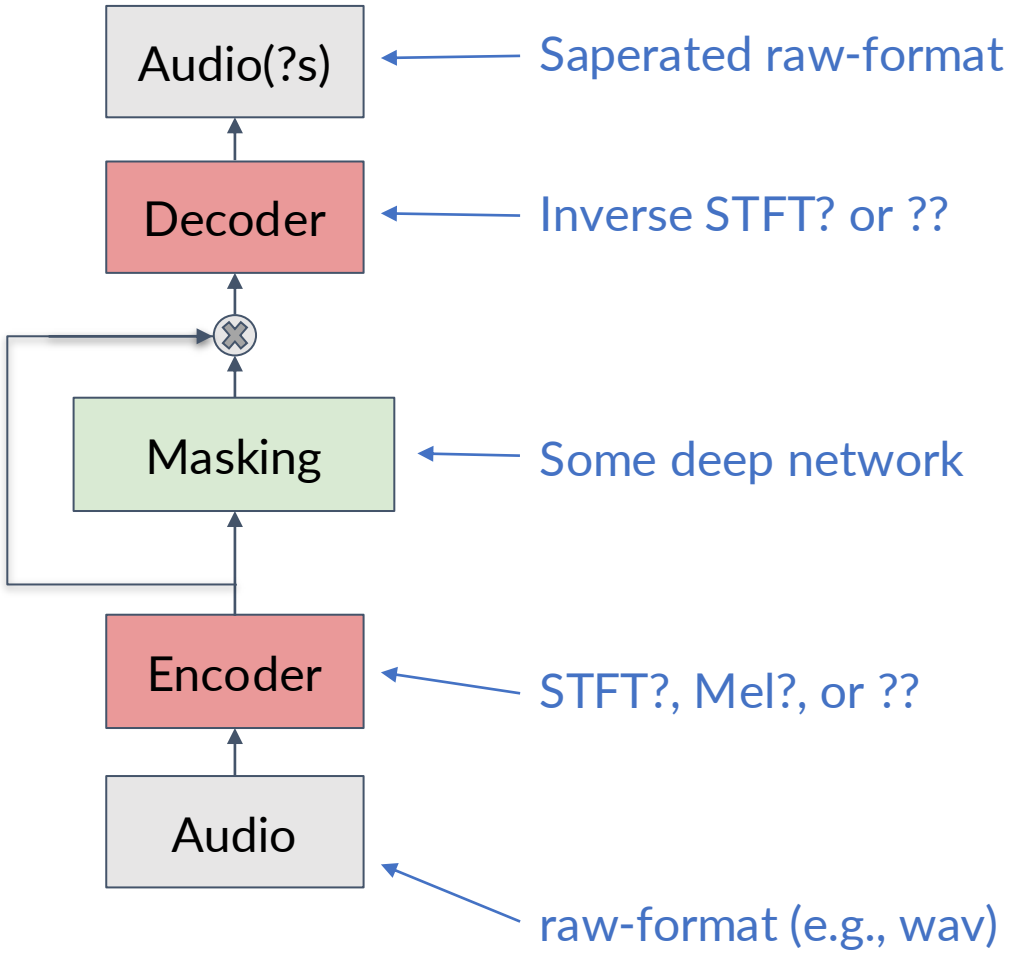
Blind Source Separation

- **Goal**: extract K sources from the noisy mixture **w/o (or with very little) information about the mixing process**

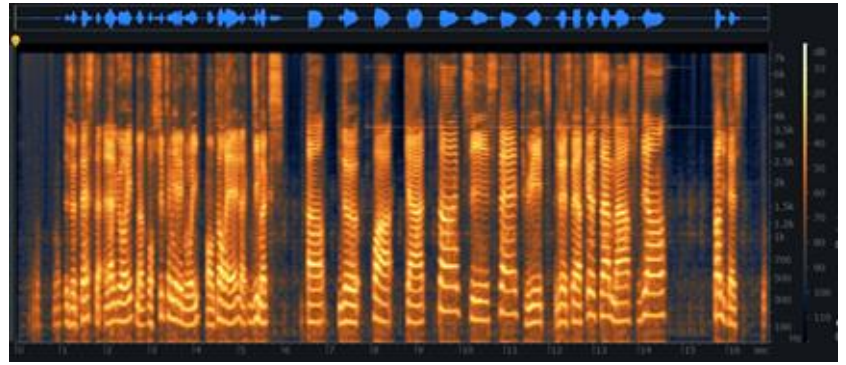


- We know denoising and specific guided separation, how can we apply DL here?

Encoder-Separation-Decoder (ESD)

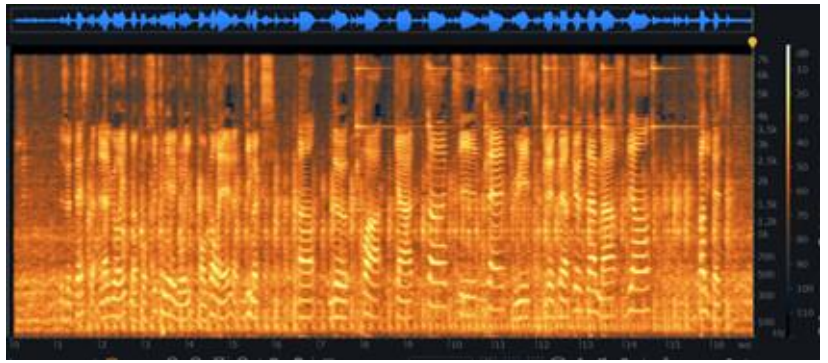


Denoised Alex



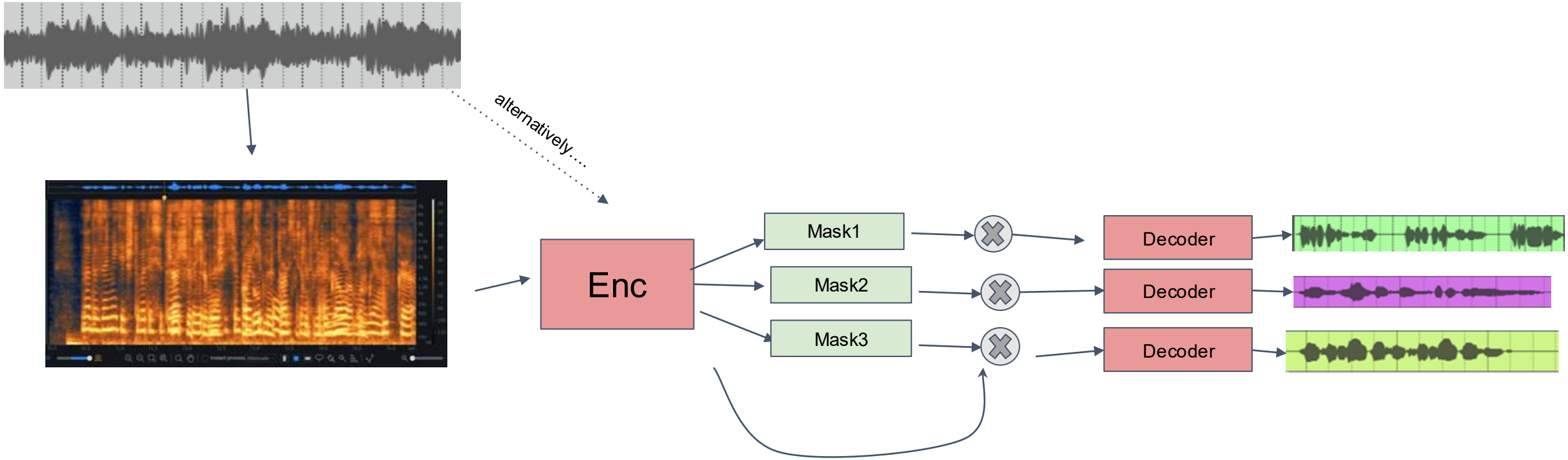
Denoising

Alex & noise



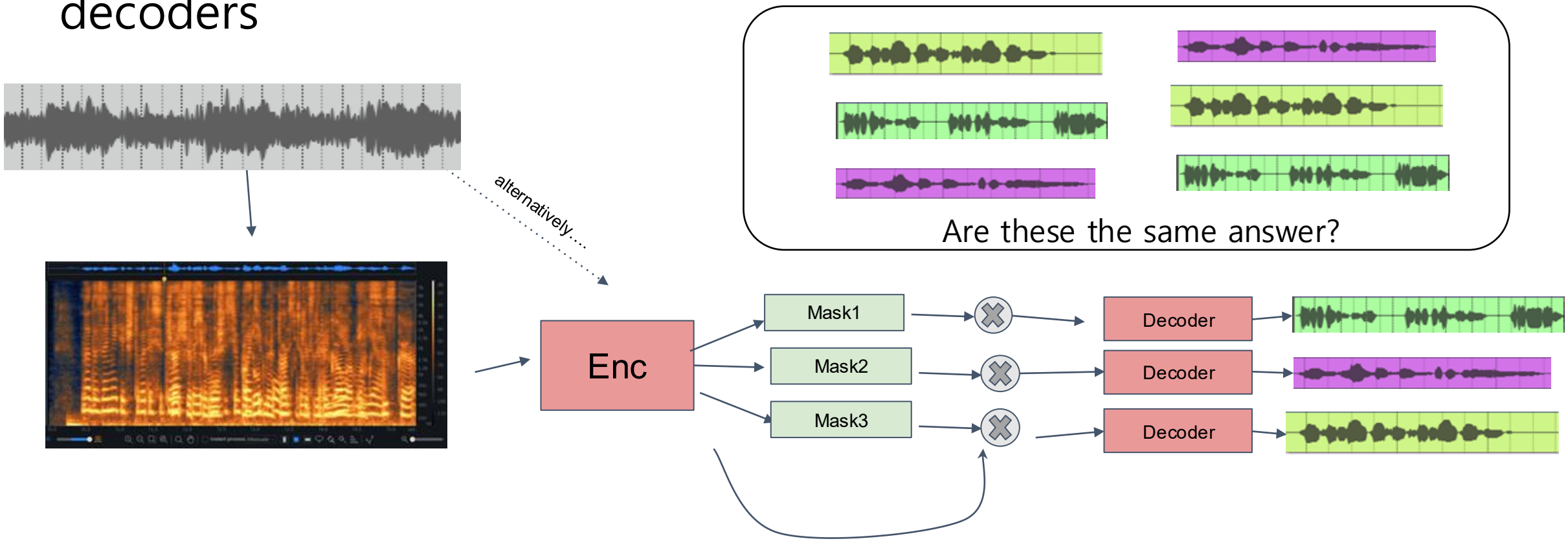
Blind Source Separation – First Idea

- **Goal**: just use Encoder-Separation-Decoder (ESD) with several decoders



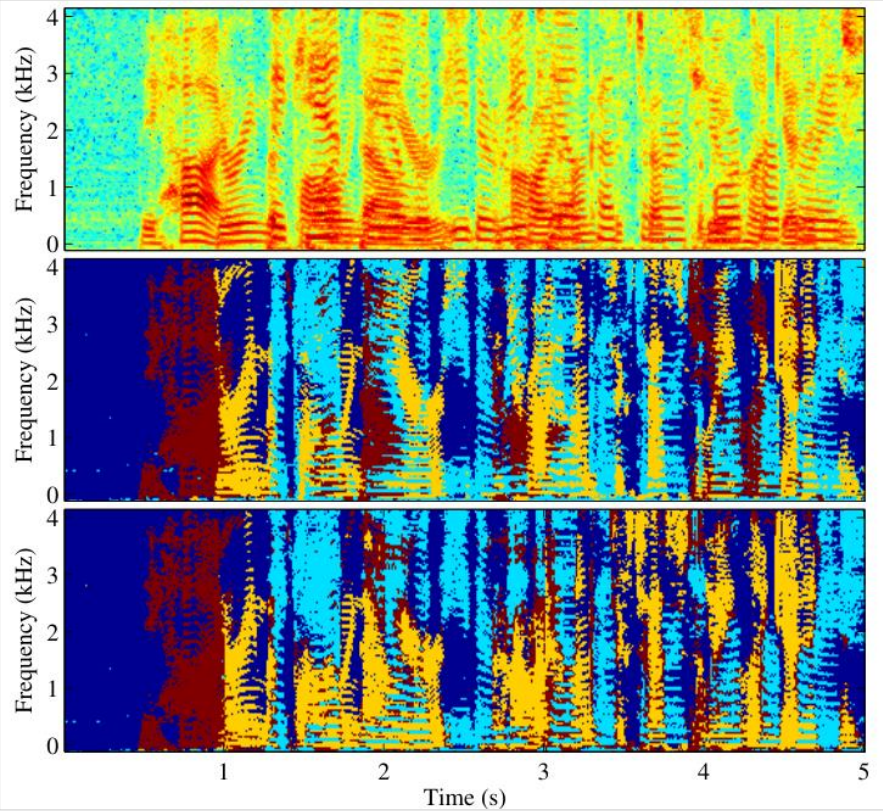
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- **Goal**: just use Encoder-Separation-Decoder (ESD) with several decoders



Before PIT – Deep Clustering (DPCL)

- **Deep Clustering [2016]**: solve the permutation problem by clustering, not by fixed output slots



Hershey et al. 2016, Fig. 1 – mixture / ideal mask / DPCL mask (3 spk)

Training: $L = \left\| VV^{\mathrm{T}} - YY^{\mathrm{T}} \right\|_F^2$

VV^{T} / YY^{T}: affinity – "same speaker?" → **no output order**

(Y: one-hot speaker label of each T-F bin)

Inference: each T-F bin → embedding V → k-means (K clusters) → binary masks → iSTFT

Pros

- Permutation-free by design
- K chosen at inference → 2-spk model separates 3 spk (left figure)

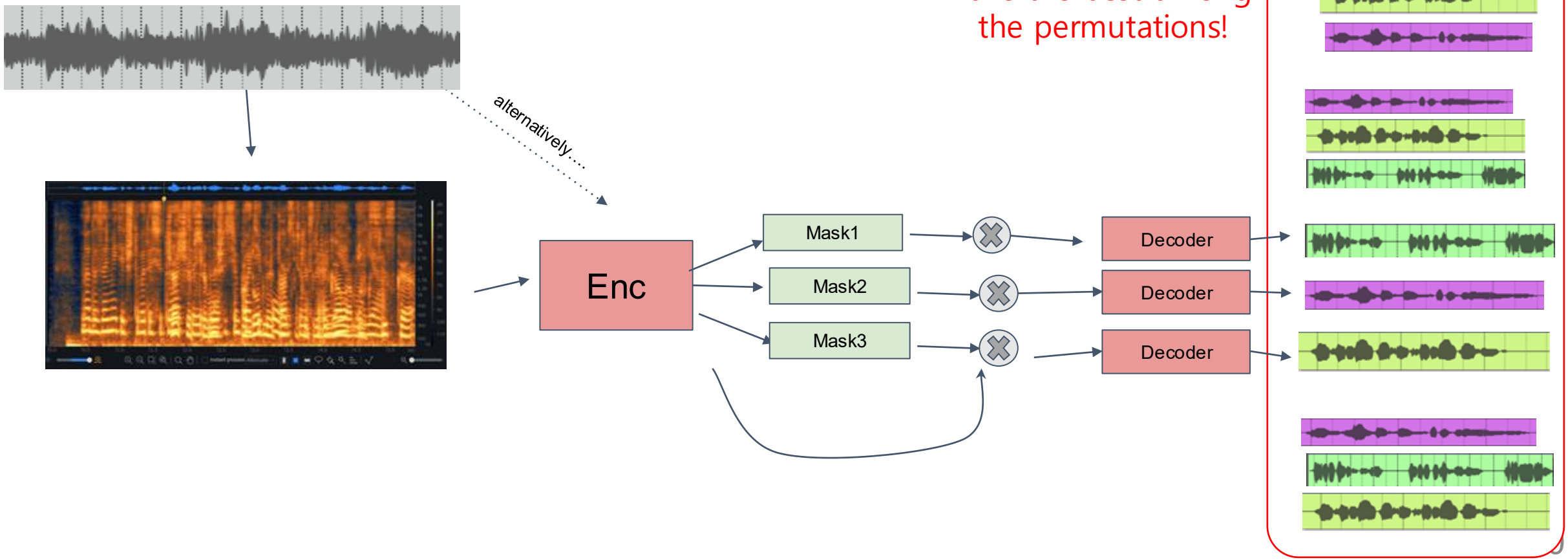
Cons → PIT

- k-means not trained end-to-end
- Binary magnitude mask, mixture phase

Hershey et al., "Deep clustering: Discriminative embeddings for segmentation and separation," ICASSP 2016; DPCL++: Isik et al., Interspeech 2016

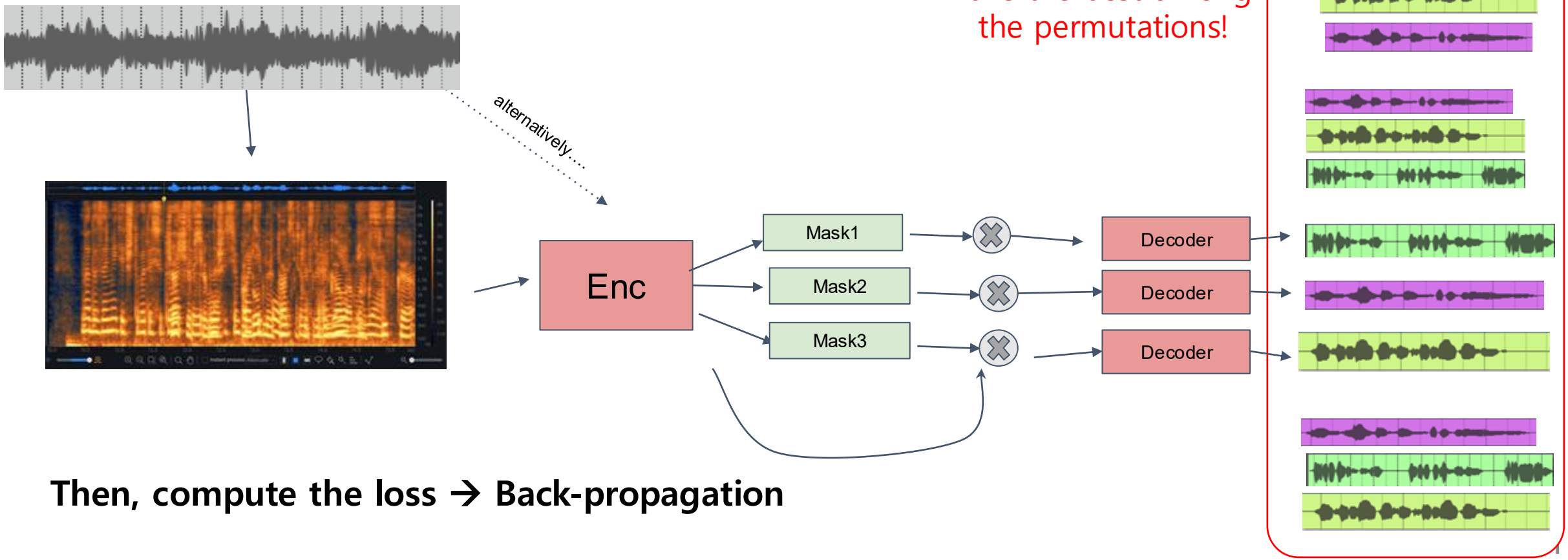
Blind Source Separation – PIT

- **Idea**: Permutation-Invariant Training (PIT), take the best loss of all permutations of predictions



Blind Source Separation – PIT

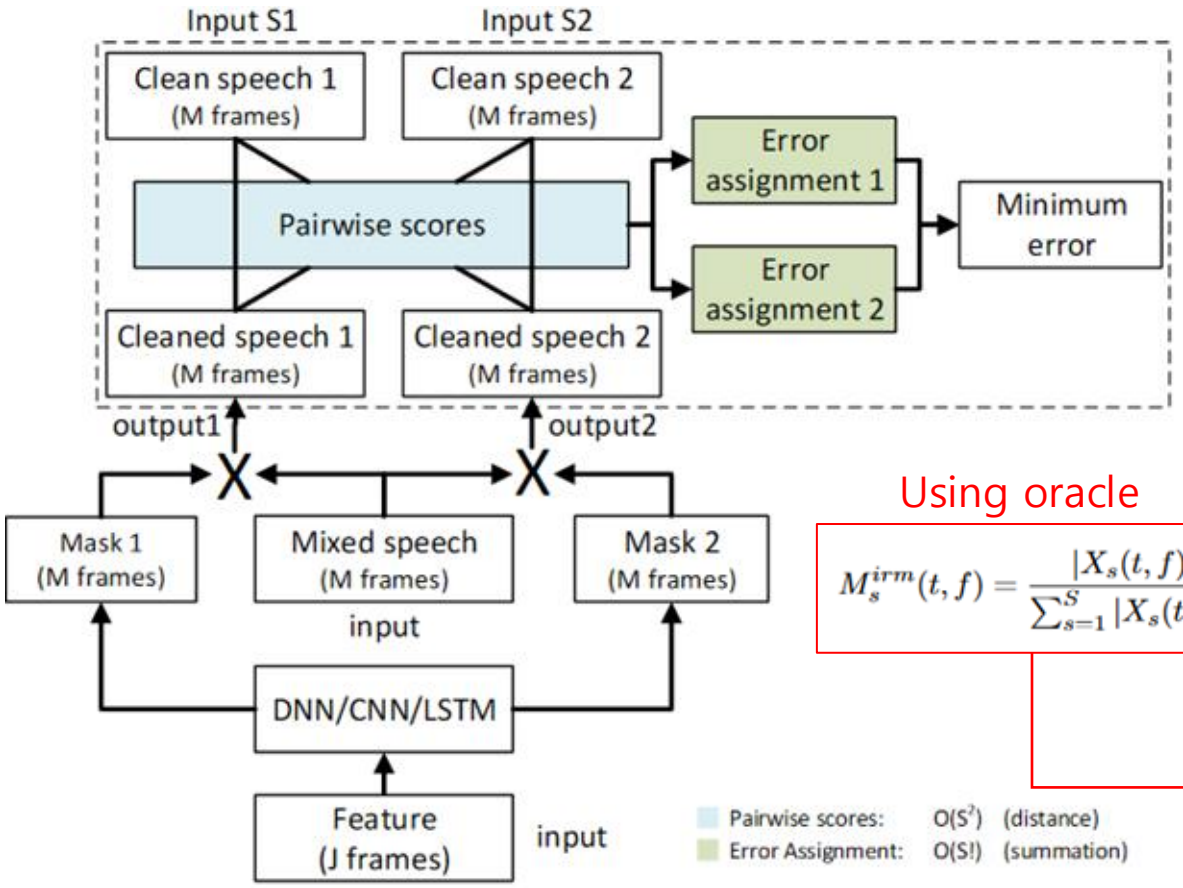
- **Idea**: Permutation-Invariant Training (PIT), take the best loss of all permutations of predictions



Then, compute the loss → Back-propagation

Blind Source Separation – PIT

- **PIT[2017] is the first working DL baseline without clustering**



Using oracle

$$M_s^{irm}(t,f) = \frac{|X_s(t,f)|}{\sum_{s=1}^{S}|X_s(t,f)|}$$

SDR IMPROVEMENTS (DB) FOR DIFFERENT SEPARATION METHODS ON THE WSJ0-2MIX DATASET USING UPIT.

Open condition (본적 없는 화자)

Method	Mask Type	Activation Function	Opt. Assign. CC	Opt. Assign. OC	Def. Assign. CC	Def. Assign. OC
uPIT-BLSTM	AM	softmax	**10.4**	**10.3**	**9.0**	**8.7**
uPIT-BLSTM	AM	sigmoid	8.3	8.3	7.1	7.2
uPIT-BLSTM	AM	ReLU	9.9	9.9	8.7	8.6
uPIT-BLSTM	AM	Tanh	8.5	8.6	7.5	7.5
uPIT-BLSTM	PSM	softmax	10.3	10.2	9.1	9.0
uPIT-BLSTM	PSM	sigmoid	10.5	10.4	9.2	9.1
uPIT-BLSTM	PSM	ReLU	**10.9**	**10.8**	**9.4**	**9.4**
uPIT-BLSTM	PSM	Tanh	10.4	10.3	9.0	8.9
uPIT-BLSTM	NPSM	softmax	8.7	8.6	7.5	7.3
uPIT-BLSTM	NPSM	sigmoid	**10.6**	**10.6**	**9.4**	**9.3**
uPIT-BLSTM	NPSM	ReLU	8.8	8.8	7.6	7.6
uPIT-BLSTM	NPSM	Tanh	10.1	10.0	8.9	8.8
uPIT-LSTM	PSM	ReLU	**9.8**	**9.8**	7.0	**7.0**
uPIT-LSTM	PSM	sigmoid	**9.8**	9.6	**7.1**	6.9
uPIT-LSTM	NPSM	ReLU	**9.8**	**9.8**	**7.1**	**7.0**
uPIT-LSTM	NPSM	sigmoid	9.2	9.2	6.8	6.8
PIT-BLSTM	PSM	ReLU	**11.7**	**11.7**	**-1.7**	-1.9
PIT-BLSTM	PSM	sigmoid	**11.7**	**11.7**	**-1.7**	**-1.7**
PIT-BLSTM	NPSM	ReLU	**11.7**	**11.7**	**-1.7**	-1.8
PIT-BLSTM	NPSM	sigmoid	11.6	11.6	-1.6	**-1.7**
IRM	-	-	12.4	12.7	12.4	12.7
IPSM	-	-	14.9	15.1	14.9	15.1

w/ PIT w/o PIT

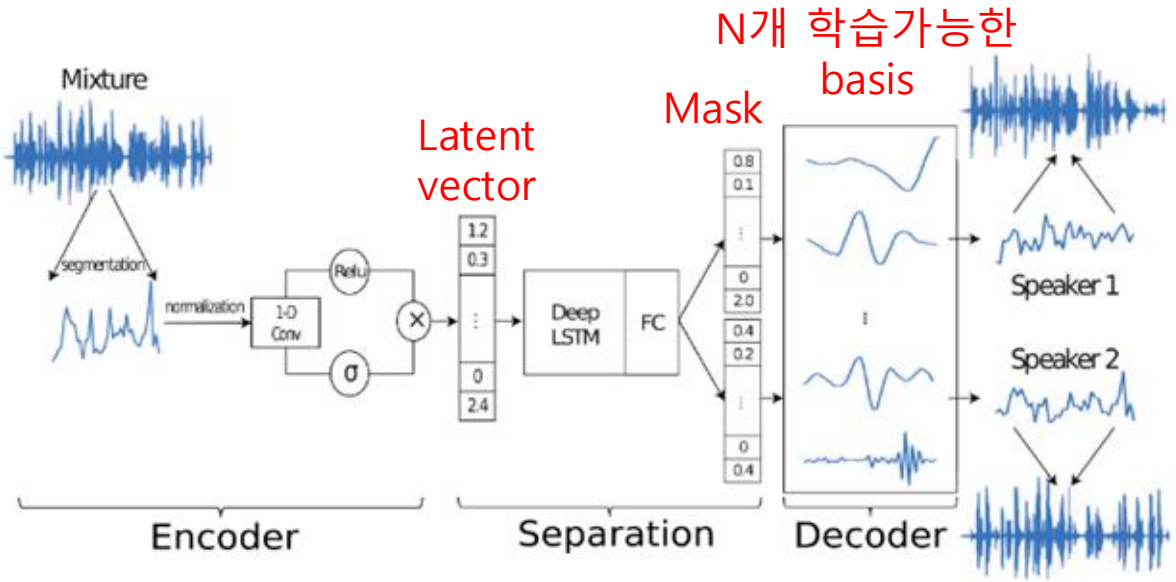
Blind Source Separation – RNN-based

- **TasNet (2017): solve blind speaker separation (2 speakers tried)**

  - **1D CNN** Encoder
  - **Heavy** (1024hid) 4-layer **LSTM**s as Separator
  - **FC**(!) decoders each of 1024 units
  - Short segments of audio (5ms at 8kHz SR) as input
  - **PIT** used

  Trained and evaluated on WSJ0-2mix

  One of the first serious DNN baselines together with PIT



Method	Causal	SI-SNRi	SDRi
uPIT-LSTM [4]	✓	–	7.0
TasNet-LSTM	✓	7.7	**8.0**
DPCL++ [3]	×	**10.8**	–
DANet [5]	×	10.5	–
uPIT-BLSTM-ST [4]	×	–	10.0
TasNet-BLSTM	×	**10.8**	**11.1**

25M parameters

SDRi: improvement over mix, SDR around 7-8(BLSTM)

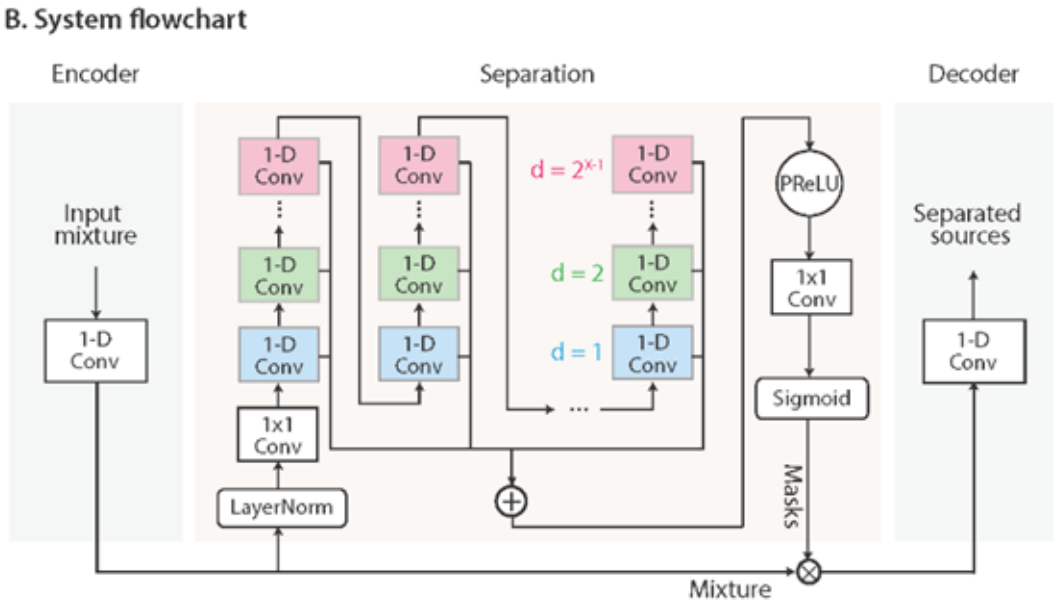
Blind Source Separation – CNN-based

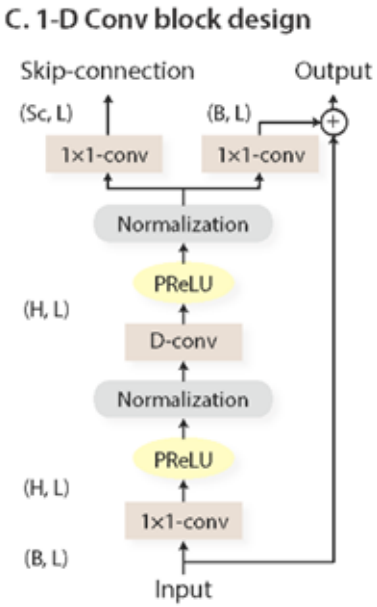
- **ConvTasNet (2018):** solve blind speaker separation (2 speakers tried)

  - **1D CNN** Encoder
  - **TCN (Temporal Convolutional Network) ResNet-like structure** as Separator
  - Arbitrary-length audio as input
  - **PIT** used

Trained and evaluated on WSJ0-2mix

The second serious DNN baseline together with PIT and TasNet





Blind Source Separation – CNN-based

- **ConvTasNet (2018):** solve blind speaker separation (2 speakers tried)

- **1D CNN** Encoder
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The second serious DNN baseline together with PIT and TasNet

Method	Model size	Causal	SI-SNRi (dB)	SDRi (dB)
DPCL++ [5]	13.6M	×	10.8	–
uPIT-BLSTM-ST [7]	92.7M	×	–	10.0
DANet [8]	9.1M	×	10.5	–
ADANet [9]	9.1M	×	10.4	10.8
cuPIT-Grid-RD [50]	47.2M	×	–	10.2
CBLDNN-GAT [12]	39.5M	×	–	11.0
Chimera++ [10]	32.9M	×	11.5	12.0
WA-MISI-5 [11]	32.9M	×	12.6	13.1
BLSTM-TasNet [26]	23.6M	×	13.2	13.6
Conv-TasNet-gLN	**5.1M**	×	**15.3**	**15.6**
uPIT-LSTM [7]	46.3M	✓	–	7.0
LSTM-TasNet [26]	32.0M	✓	**10.8**	**11.2**
Conv-TasNet-cLN	**5.1M**	✓	10.6	11.0
IRM	–	–	12.2	12.6
IBM	–	–	13.0	13.5
WFM	–	–	13.4	13.8

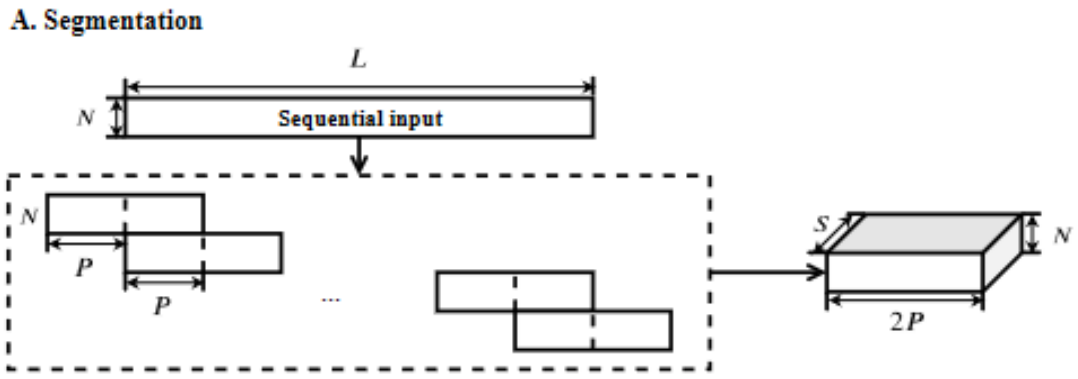
Blind Source Separation – RNN-based

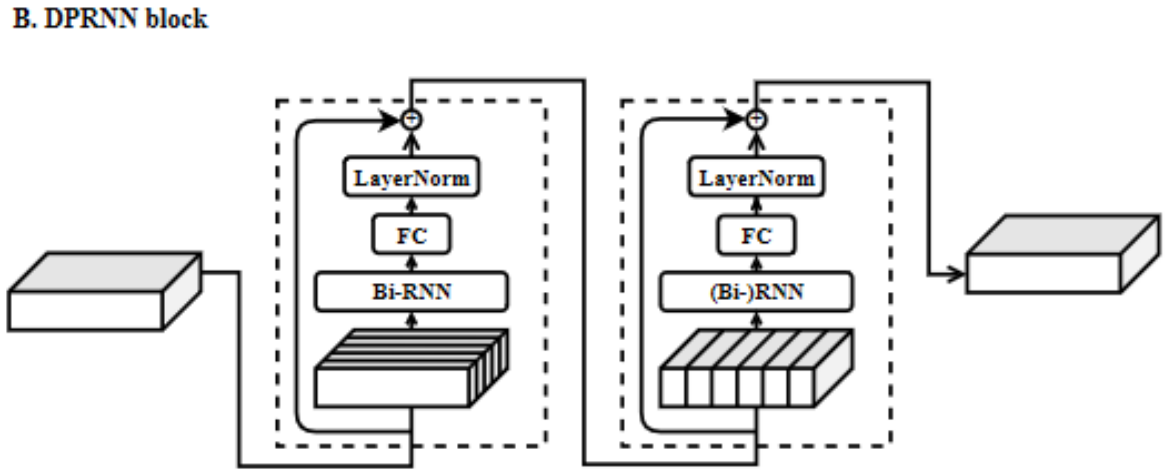
- **Dual-Path RNN (2020):** solve blind speaker separation (2 speakers tried)

  - **Dual-Path RNN**
  - **Light:** 2.6M parameters
  - **Streaming**-ready
  - Short overlapping chunks of audio (2ms at 16kHz SR) as input

  Application: speech enhancement, separation with known number of speakers…

  SI-SNRi: WSJ02-mix ~18



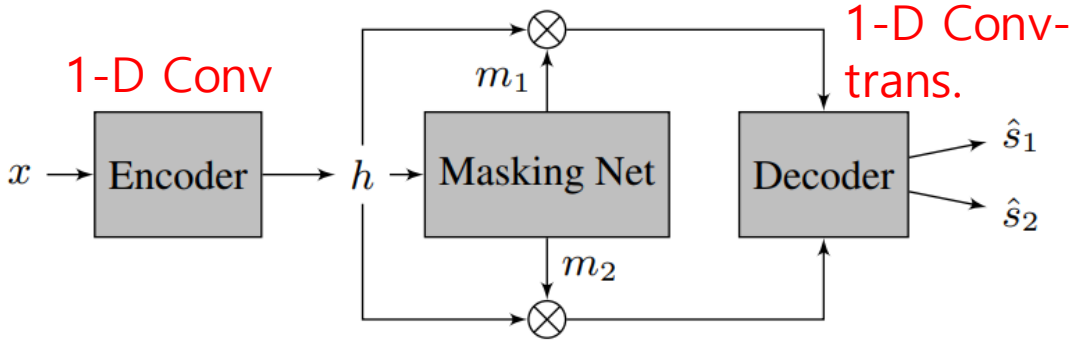


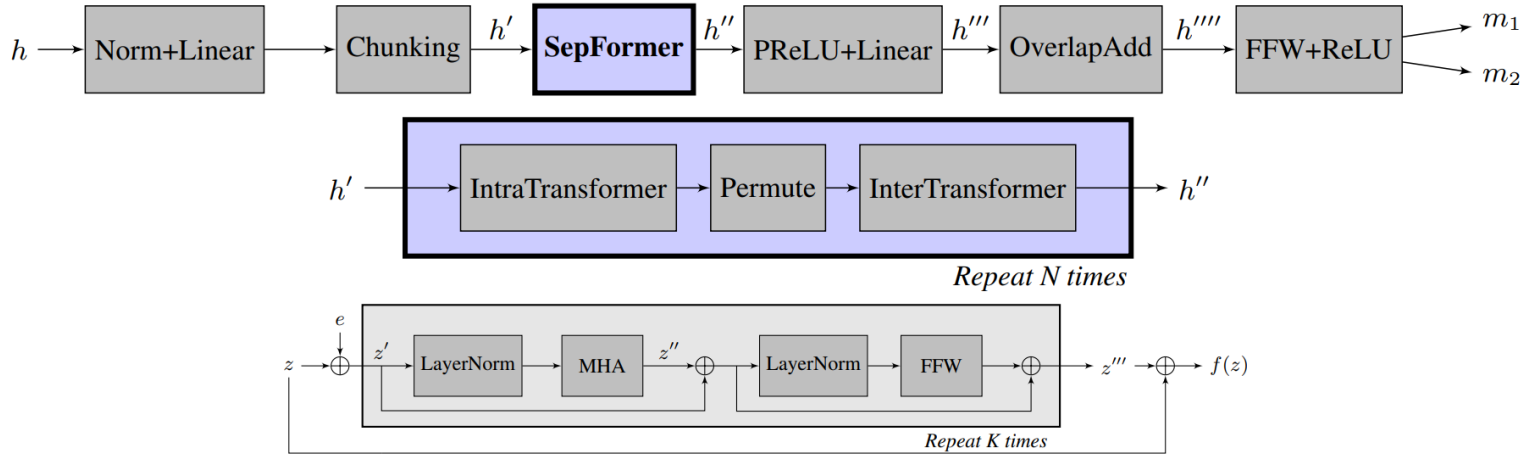
Blind Source Separation – Transformer-based

- **SepFormer (2021):** solve blind speaker separation (2 speakers tried)

- **Transformer**-based dual-path architecture
- **Self-Attention** layers replace TCN blocks in separator
- Handles **long-term temporal dependencies** efficiently
- Fully convolutional Encoder/Decoder in time domain
- **PIT** (Permutation Invariant Training) used

Trained and evaluated on WSJ0-2mix and WHAM! datasets





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Table 1. Best results on the WSJ0-2mix dataset (test-set). DM stands for dynamic mixing.

Model	SI-SNRi	SDRi	# Param	Stride
Tasnet [27]	10.8	11.1	n.a	20
SignPredictionNet [28]	15.3	15.6	55.2M	8
ConvTasnet [15]	15.3	15.6	5.1M	10
Two-Step CTN [29]	16.1	n.a.	8.6M	10
DeepCASA [18]	17.7	18.0	12.8M	1
FurcaNeXt [19]	n.a.	18.4	51.4M	n.a.
DualPathRNN [17]	18.8	19.0	2.6M	1
sudo rm -rf [21]	18.9	n.a.	2.6M	10
VSUNOS [20]	20.1	20.4	7.5M	2
DPTNet* [22]	20.2	20.6	2.6M	1
Wavesplit** [23]	21.0	21.2	29M	1
Wavesplit** + DM [23]	22.2	22.3	29M	1
SepFormer	20.4	20.5	26M	8
SepFormer + DM	22.3	22.4	26M	8

Separation Lineage at a Glance

Model	Year	Encoder / Domain	Separator	Params	SI-SNRi (dB)	Causal variant
DPCL++	2016	STFT magnitude	BLSTM + k-means	–	10.8	✗
uPIT-BLSTM	2017	STFT magnitude	BLSTM	–	≈10 (SDRi)	✓ (LSTM)
TasNet-BLSTM	2018	Learned 1-D conv	Deep (B)LSTM	23.6M	10.8	✓ (LSTM)
Conv-TasNet	2019	Learned 1-D conv	TCN (dilated conv)	5.1M	15.3	✓ (cLN)
DPRNN	2020	Learned 1-D conv	Dual-path BLSTM	2.6M	18.8	–
SepFormer	2021	Learned 1-D conv	Dual-path Transformer	26M	20.4 / 22.3 (DM)	✗

Common frame

- Encoder → Separator (masks) → Decoder
- SI-SNR loss + utterance-level PIT
- Only the separator changes: RNN → CNN → Transformer

Reference line

- Oracle STFT masks (IRM/IBM/WFM): 12–13 dB
- Conv-TasNet (2019) is the first to exceed them
- DM = dynamic mixing (data augmentation)

WSJ0-2mix (8 kHz, 2 speakers), numbers as reported in the original papers. uPIT row reports SDRi.

Recent Trend ① – Back to T-F Domain

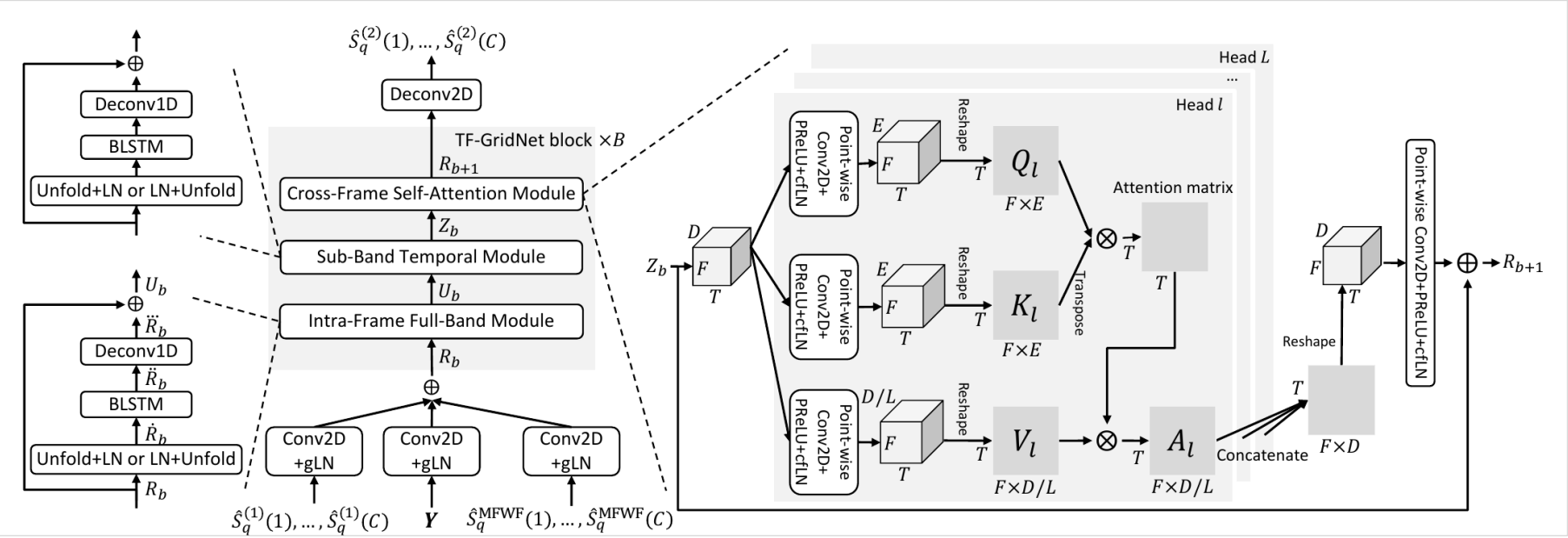
- **2019: time-domain beat oracle STFT masks → 2023: TF-GridNet returns to the STFT (23.5 dB)**

What changed?

- **Before:** mask on $|X|$, mixture phase: $\hat{S} = M \odot |X| e^{j\angle X}$
- **Now:** predict $\mathrm{Re}(\hat{S}), \mathrm{Im}(\hat{S})$ directly → phase estimated too

The limit was the magnitude mask, not the STFT

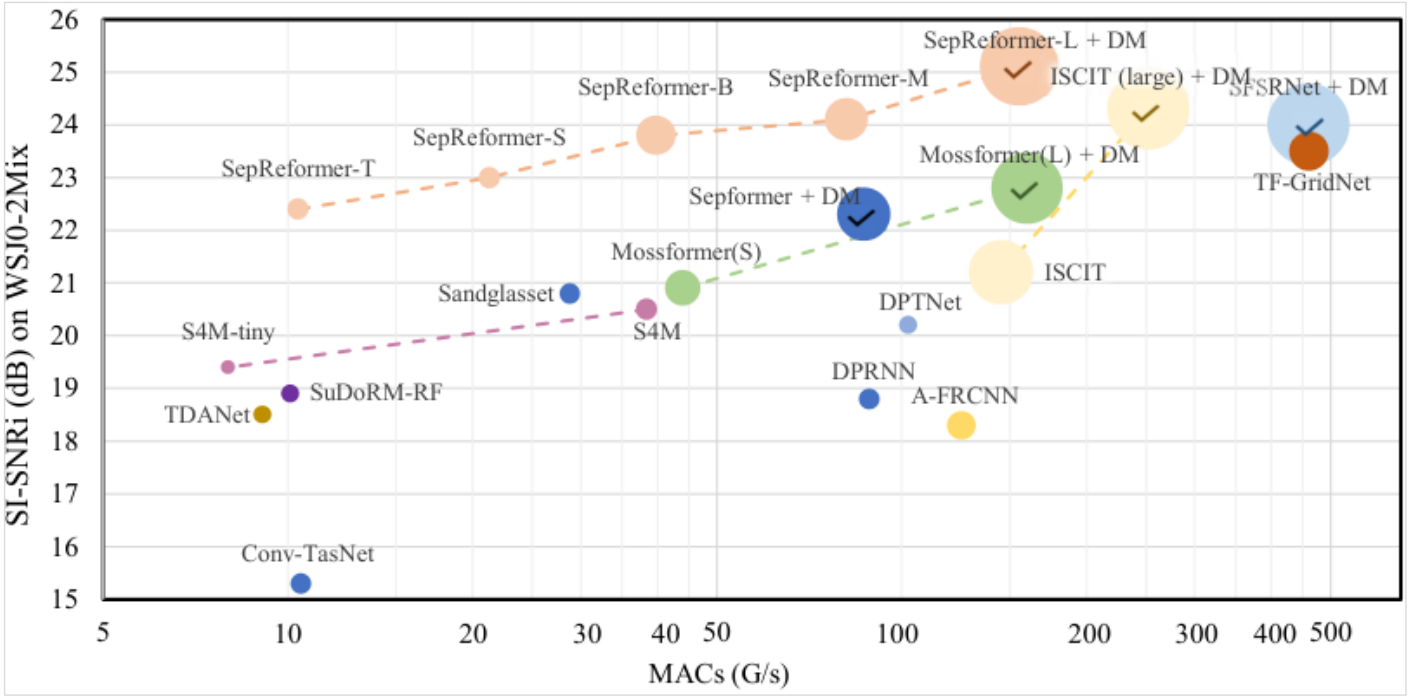
→ Lecture 6: phase problem



TF-GridNet block (Wang et al. 2023, Fig. 2): intra-frame full-band (across F) → sub-band temporal (across T) → cross-frame self-attention

Wang et al., "TF-GridNet: Integrating full- and sub-band modeling for speech separation," IEEE/ACM TASLP 2023

Recent Trend ② – Efficiency & Realism



SI-SNRi on WSJ0-2mix vs. MACs (Shin et al. 2024, Fig. 5); circle size = params, ✓ = dynamic mixing

≈25 dB → WSJ0-2mix is nearly saturated

New question: smaller, faster, and robust in the wild

TIGER (ICLR 2025)

- Band-split + multi-scale attention
- ~94% fewer params, ~95% fewer MACs than TF-GridNet
- EchoSet: realistic reverb & noise → better real-world generalization

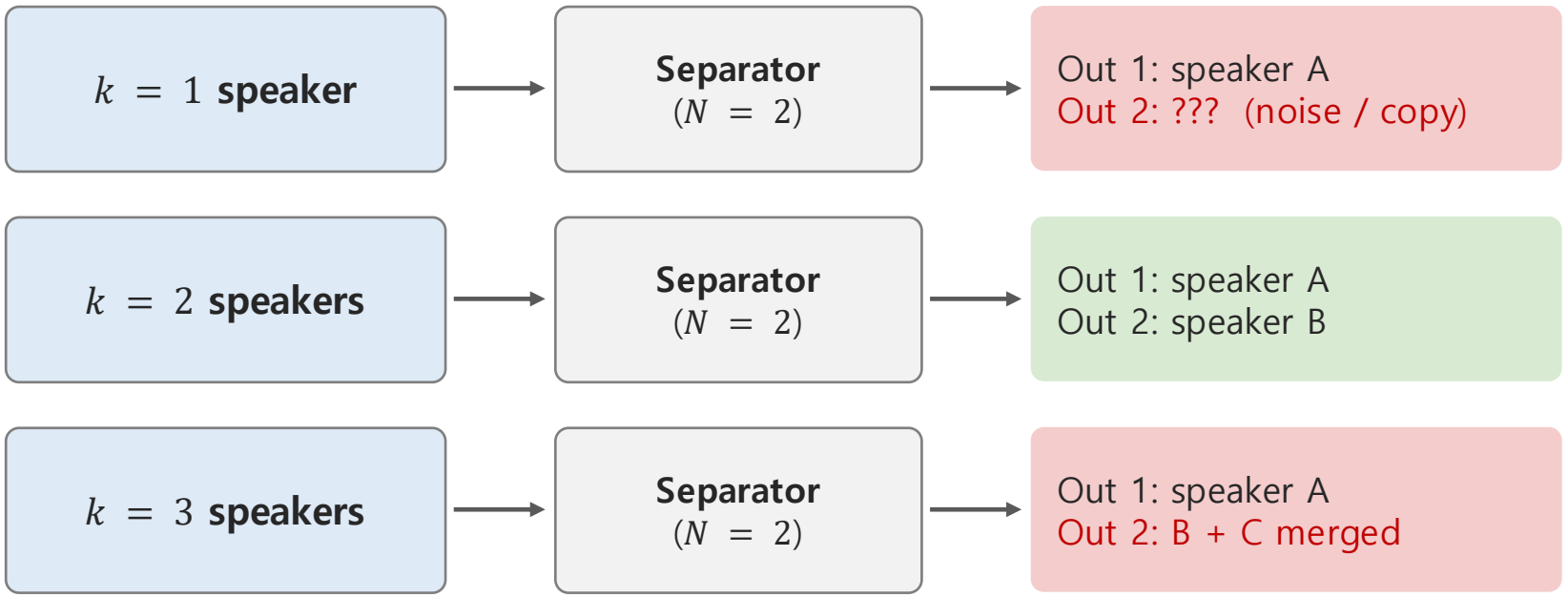
SepMamba (2024)

- Mamba (state-space) blocks instead of self-attention
- Linear-time in sequence length → long audio, low memory

Shin et al., SepReformer, NeurIPS 2024. Xu et al., TIGER, ICLR 2025. Avenstrup et al., SepMamba, arXiv 2410.20997

Open Problem ① – Speaker Count

- **"Blind (any k)"?** Every model so far has a fixed number of outputs N (WSJ0-2mix → $N = 2$)



In practice

- Meetings, robots, smart speakers: k is unknown and changes over time
- Two sub-problems: counting + separation
- PIT cost grows as $k!$ (or k^2 with Hungarian)

Benchmarks: WSJ0-{2,3,4,5}mix, Libri2Mix / Libri3Mix

Speaker Count – Three Approaches

(a) Max outputs + silence

- Train $N_{\max}$ outputs; unused ones target silence
- Drop low-energy outputs at inference

+ Simple – $N_{\max}$ still fixed

(b) Recursive extraction

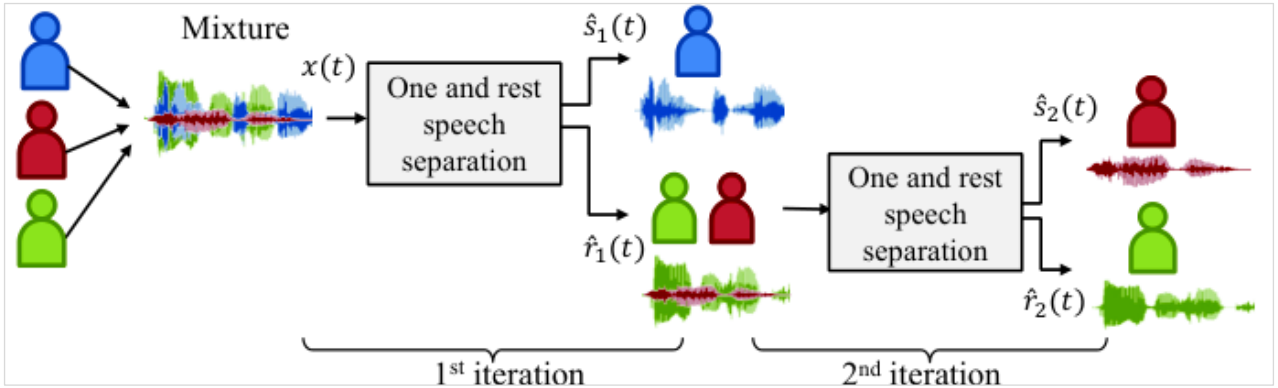
- OR-PIT [2019]: one speaker + rest per step
- Repeat on the rest until it is empty

+ Any k, one model – Error build-up

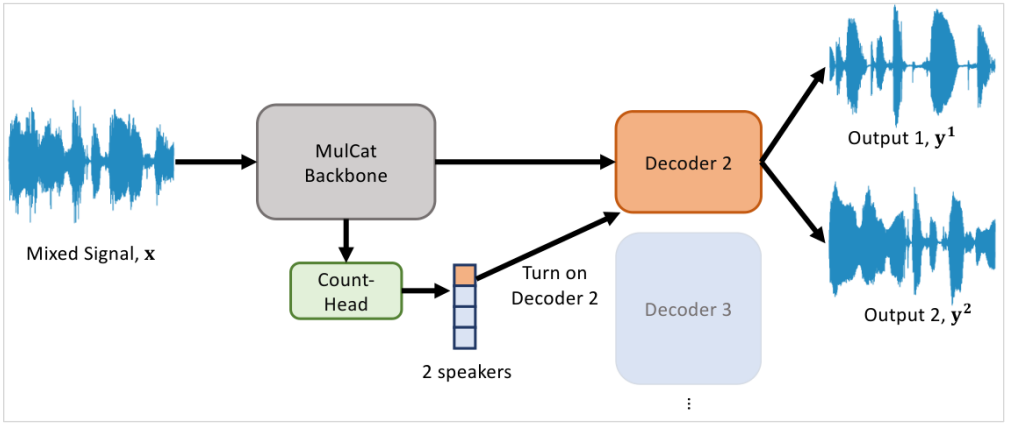
(c) Count + separate

- Multi-decoder DPRNN [2021]: count head picks the decoder
- Nachmani [2020]: one model per k; SepTDA [2024]: attractors

+ Joint counting – Count errors are costly



(b) Recursive separation with OR-PIT, N = 3 (Takahashi et al. 2019, Fig. 1)



(c) Count head selects decoder 2 (Zhu et al. 2021, Fig. 1)

Takahashi et al., Interspeech 2019; Zhu et al., ICASSP 2021; Nachmani et al., ICML 2020; Lee et al. (SepTDA), ICASSP 2024

Open Problem ② – Real-World Gap

WSJ0-2mix (2016)
Clean read speech, anechoic, 8 kHz

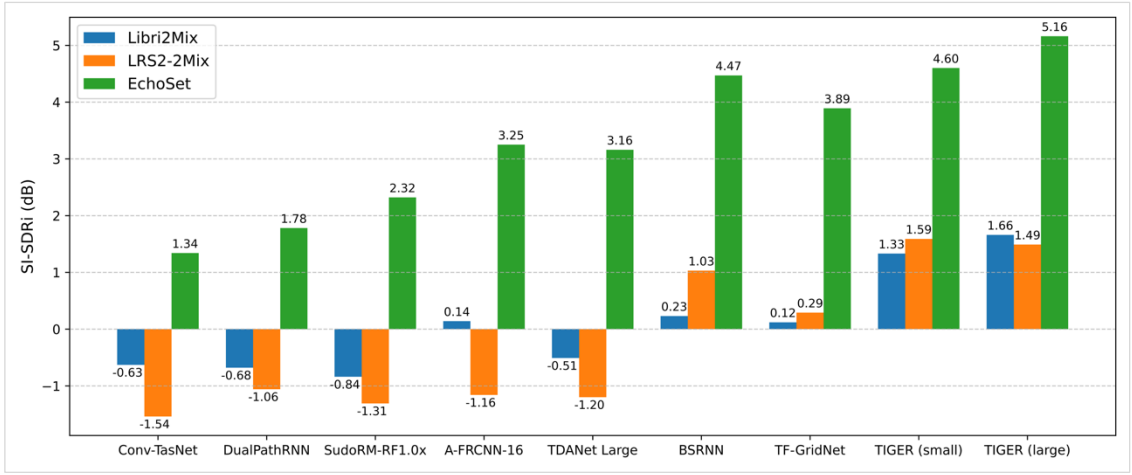
WHAM! (2019)
+ Real ambient noise

WHAMR! (2020)
+ Reverberation

LibriMix (2020)
More speakers & hours, 16 kHz, noisy

REAL-M (2022) · EchoSet (2025)
Real / realistic acoustics

More realistic ↓ → SI-SNRi ↓



SI-SDRi on real-world recordings by training set (Xu et al. 2025, Fig. 4)

What the figure shows

- Trained on Libri2Mix / LRS2-2Mix: near 0 or negative on real data
- Trained on EchoSet (realistic acoustics): clearly positive
- **→ training-data realism matters as much as the model**

Wichern 2019 (WHAM!); Maciejewski 2020 (WHAMR!); Cosentino 2020 (LibriMix); Subakan 2022 (REAL-M); Xu 2025 (EchoSet, TIGER)

Real-World Training – DM & MixIT

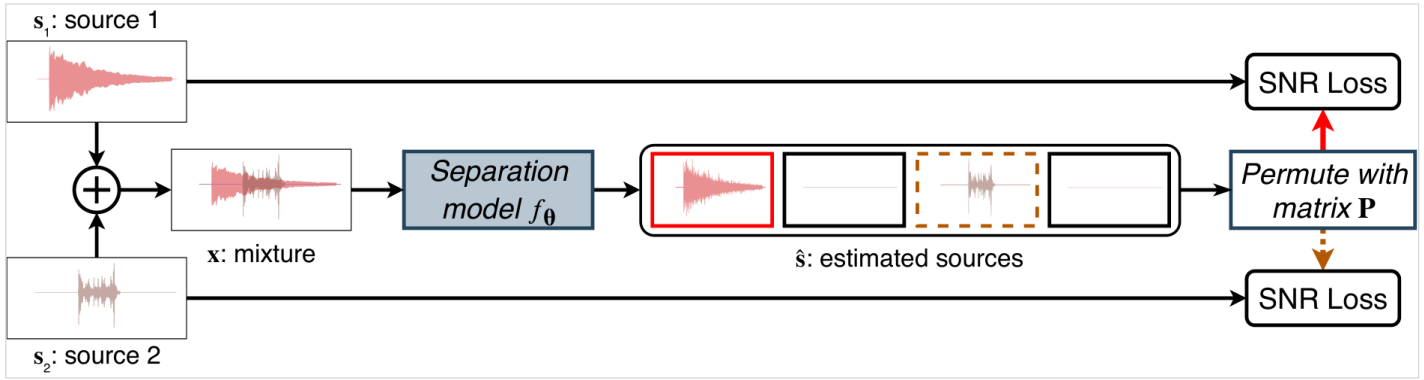
Dynamic Mixing (DM)

- Re-mix clean sources on the fly (random pairs, gains, shifts)
- Effectively infinite mixtures → less overfitting
- **SepFormer: 20.4 → 22.3 dB**

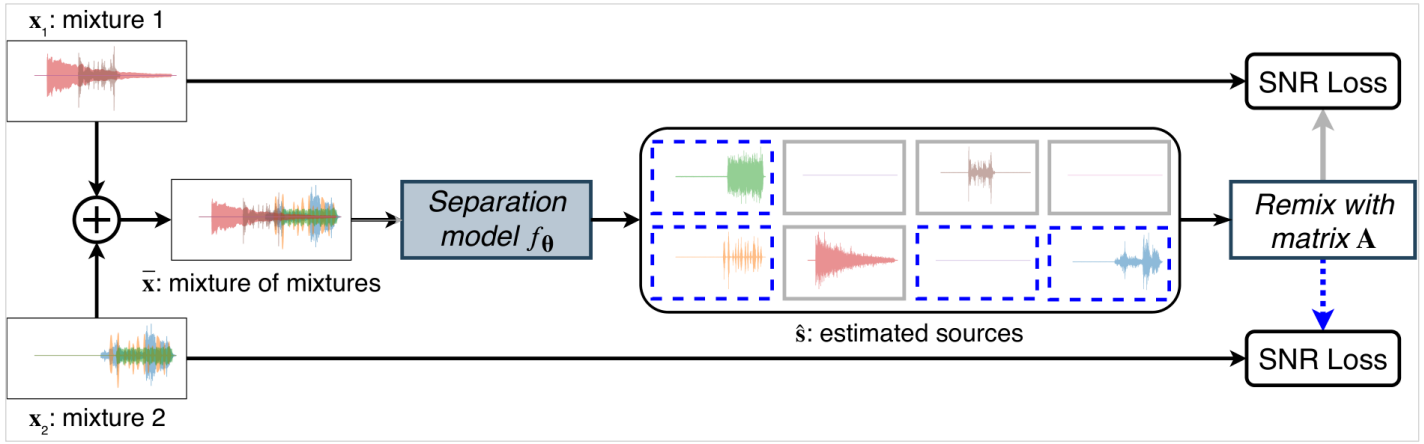
MixIT [2020]

- Input: mixture of mixtures $x_1 + x_2$
- Remix outputs to rebuild x_1 and x_2
- **Unsupervised:** no clean sources needed
- Limit: over-separation

(a) PIT – supervised: permute outputs to match sources



(b) MixIT – unsupervised: remix outputs to match mixtures

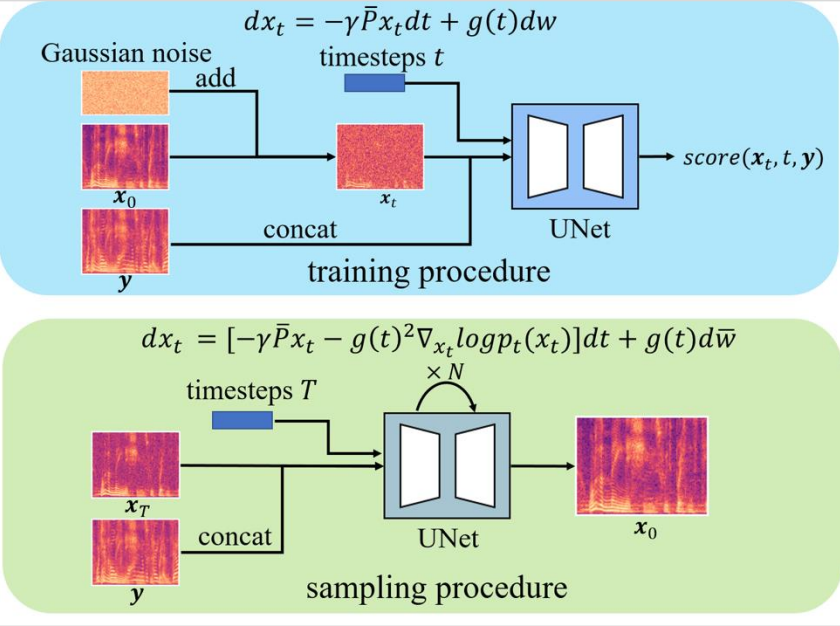


Subakan et al., SepFormer, ICASSP 2021 (DM). Wisdom et al., "Unsupervised sound separation using mixture invariant training," NeurIPS 2020, Fig. 1

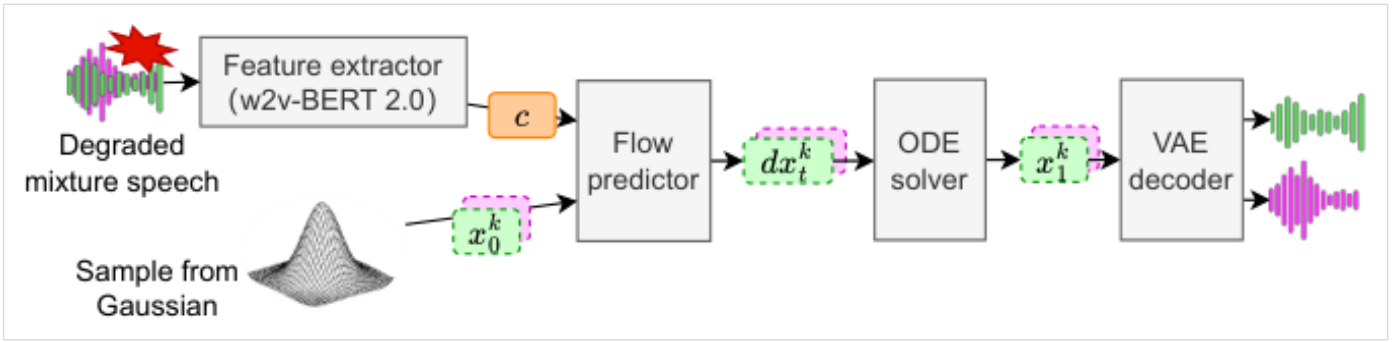
Recent Trend ③ – Generative Models

Discriminative (so far)
- mixture → mask / estimate: one best guess
- Regression to the mean → muffled / leaky

Generative
- Learn $p(s_1, \ldots, s_K \mid x)$, then sample
- Diffusion / score-based, flow matching



Diffusion training & sampling for separation (EDSep 2025, Fig. 1)



Flow matching in VAE latent space (Geneses 2026, Fig. 1)

\+ Natural output, restores lost parts

– Slow sampling, may hallucinate

Scheibler et al. (DiffSep), ICASSP 2023; EDSep, arXiv 2501.15965; Geneses, arXiv 2601.18456